CONFOEDERATIO,
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DEMOGRAPHIC AND SETTLEMENT HISTORY OF THE TSARDOM OF RUSSIA, BALTIC PROVINCES, LITHUANIAN-POLISH RUS, AND SIBERIAN COLONISATION.

A Regional Literature Review of Population and Settlement (1400AD - 1721AD).

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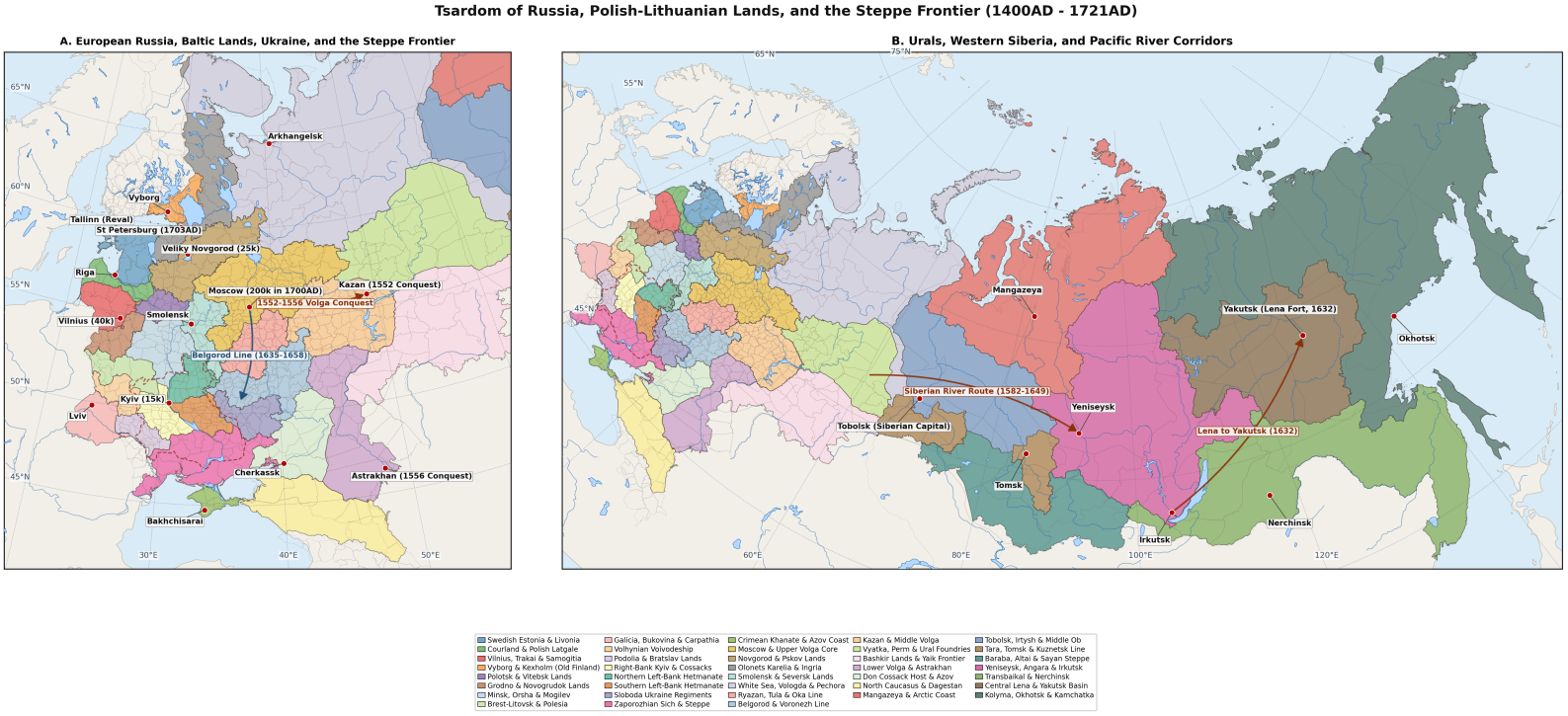


Figure 1. Historical regions and major urban centres across the Baltic provinces, Lithuania, Ukraine, the Tsardom of Russia, and Siberia (1400AD - 1721AD).

Population Estimates and Historical Regions (1400AD - 1721AD)

Between 1400AD and the Treaty of Nystad at the conclusion of the Great Northern War in 1721AD, the Northern Eurasian, East Slavic, and Eastern Baltic landmass underwent a fundamental demographic, institutional, and spatial transformation. In 1400AD, the combined population across the thirty-eight historical regions spanning the Eastern Baltic confederations, the Grand Duchy of Lithuania, Vyborg-Kexholm Karelia, Crown and Cossack Ukraine, Crimea, the Tsardom of Russia, the North Caucasus, and Siberia stood at 11 355k (11,36M). At that late medieval benchmark, regional population comprised 935k across Terra Mariana and Livonia, Courland-Inflanty, core Lithuania, and Vyborg-Kexholm (Regions 1-4), 950k across the four Belarusian voivodeships of the Grand Duchy of Lithuania (Regions 5-8), 2 510k across the nine Ukrainian and Crimean-Pontic historical regions (Regions

9-17), and 6 960k across the twenty-one Russian, Volga-Ural, Steppe, Caucasian, and Siberian regions (Regions 18-38, divided between 6 660k in European Russia, the Steppe, and the North Caucasus across Regions 18-30 and 300k across the eight Siberian functional zones Regions 31-38) [1][2][3][4][5][6]. Over the subsequent three centuries, total domain population doubled to 20 695k (20,70M) on the eve of the mid-17th-century wars in 1648AD and reached 22 830k (22,83M) at Peter I's First Poll-Tax Revision in 1720AD, comprising 1 260k in the Baltic, Lithuanian, and Old Finland provinces (Regions 1-4), 1 550k in Belarus (Regions 5-8), 5 650k in Ukraine and Crimea (Regions 9-17), and 14 370k across Russia and Siberia (Regions 18-38: 13 610k in Regions 18-30 and 760k across Regions 31-38) [7][8][9][10][11].

This long-term expansion was governed by the interaction of two opposing demographic regimes. On one side, recurrent military, climatic, and epidemic shocks repeatedly struck the western littoral and older forest cores: the Livonian War and Ivan IV's *Oprichnina* (1558-1583AD), the Huaynaputina volcanic-winter famine of 1601-1603AD and the Time of Troubles (*Smuta*, 1604-1618AD), the Russo-Polish War or Deluge (*Potop*, 1654-1667AD) across Lithuania and Belarus, *The Ruin* (*Ruina*, 1657-1687AD) in Right-Bank Ukraine, the Great Famine of 1695-1697AD (*Suur nalg* and *Suuret kuolonvuodet*) in Estland, Livland, and Old Finland, and the Great Northern War and Baltic-Lithuanian Plague of 1708-1711AD [12][13][14][15][16]. On the other side, sustained frontier colonisation transformed the Left-Bank Cossack Hetmanate, Sloboda Ukraine, the Oka-Belgorod-Simbirsk fortification lines (*Zasechnye Cherty*), the Middle Volga, the Demidov Urals, and the river-portage highways and southern fortified ploughlands of Siberia [17][18][19][20].

Eliminating anachronistic post-1991 state borders while deliberately excluding Central Asia and the South Caucasus prior to their 18th- and 19th-century annexations, this synthesis partitions the 18 119 929 km2 domain into thirty-eight historical regions constructed from Livonian and Swedish Baltic provinces, Courlandian and Polish-Lithuanian palatinates, Vyborg and Kexholm fiefs, Cossack regimental territories, pre-modern Muscovite *uezds* and frontier *razryads*, Volga-Ural *darugas*, and eight Siberian portage, fortified ploughland, steppe, and *yasak* zones (Figure 1 and Table 1). Across the Baltic, Lithuanian, and Old Finland rim (Regions 1-4), Terra Mariana and Swedish Estland and Livland (Region 1, 76 229 km2) shifted from 260k in 1400AD to 360k in 1648AD and 300k in 1720AD; the Duchy of Courland and Semigallia and Polish Inflanty (Region 2, 37 284 km2) moved from 140k to 260k and 200k; the Grand Duchy of Lithuania core of Vilnius, Trakai, and Samogitia (Region 3, 65 877 km2) expanded from 460k to 1 060k before contracting to 650k; and Vyborg Fief and Kexholm County (Region 4, 49 685 km2) rose from 75k to 110k. Across the four Belarusian voivodeships of the Grand Duchy of Lithuania (Regions 5-8), Polotsk and Vitebsk on the Western Dvina (Region 5, 32 576 km2; 240k in 1400AD to 560k in 1648AD to 370k in 1720AD), Grodno and Novogrudok in Black Ruthenia (Region 6, 48 968 km2; 280k to 740k to 490k), Minsk and the Orsha-Mogilev Dnieper corridor (Region 7, 102 278 km2; 260k to 680k to 430k), and Brest-Litovsk and Pinsk-Turov Polesia (Region 8, 62 824 km2; 170k to 520k to 260k) all peaked on the eve of the mid-17th-century wars before suffering severe wartime losses.

Across the nine Ukrainian and Crimean-Pontic regions (Regions 9-17), the Ruthenian and Belz Voivodeships of Galicia with Bukovina and Carpathian Rus (Region 9, 71 558 km2) grew from 820k in 1400AD to 1 750k in 1648AD and 1 620k in 1720AD, whereas Volhynia (Region 10, 45 176 km2; 420k to 920k to 680k), Podolia and Bratslav (Region 11, 39 620 km2; 350k to 910k to 510k), and the Right-Bank Kyiv Voivodeship and Cossack regiments (Region 12, 56 764 km2; 280k to 920k to 340k) experienced deep late-17th-century depopulation that fed rapid growth in the Northern Left-Bank Hetmanate around Chernihiv and Nizhyn (Region 13, 55 782 km2; 210k to 490k to 790k), the Southern Left-Bank Hetmanate around Pereyaslav and Poltava (Region 14, 52 082 km2; 110k to 460k to 760k), and the Sloboda Ukraine regiments around Kharkiv and Sumy (Region 15, 60 982 km2; 50k to 160k to 400k), alongside the Zaporozhian Sich and Pontic Wild Fields (Region 16, 193 859 km2; 60k to 130k to 180k) and the Crimean Khanate and Ottoman Eyalet of Kefe (Region 17, 32 002 km2; 210k to 360k to 370k).

Within the thirteen European Russian, Volga-Ural, Steppe, and North Caucasian regions (Regions 18-30),

the Moscow and Zamoskovny core (Region 18, 277 154 km2) grew from 1 950k in 1400AD to 2 850k in 1648AD and 3 720k in 1720AD, flanked by the Novgorod and Pskov Five Pyatiny (Region 19, 159 354 km2; 920k to 930k to 1 230k), Olonets Karelia and Baltic Ingria including Saint Petersburg (Region 20, 205 031 km2; 200k to 220k to 350k), the Smolensk and Seversk borderlands (Region 21, 83 068 km2; 480k to 640k to 730k), and the northern Pomorye, Vologda, Northern Dvina, and Vychegda-Pechora lands (Region 22, 1 385 055 km2; 210k to 460k to 420k). South and east of the Oka, rapid agrarian and military colonisation transformed the Ryazan, Tula, Oryol, and Great Oka Abatis Line (Region 23, 95 590 km2; 680k to 1 080k to 1 520k), the Belgorod and Sevsk Razryads (Region 24, 148 778 km2; 190k to 620k to 1 440k), the former Khanate of Kazan, Nizhny Novgorod, and the Simbirsk-Penza Line (Region 25, 237 582 km2; 780k to 1 180k to 1 760k), Great Perm, Vyatka, and the Stroganov-Demidov Urals (Region 26, 479 789 km2; 220k to 480k to 680k), the Bashkir Four Darugas and Yaik frontier (Region 27, 484 827 km2; 180k to 260k to 340k), the Lower Volga, Astrakhan, and Kalmyk Steppe (Region 28, 251 397 km2; 240k to 320k to 430k), the Don Cossack Host and Azov (Region 29, 169 534 km2; 90k to 165k to 250k), and the North Caucasus across Circassia, Kabarda, Vainakh societies, Dagestan, and Terskiy Gorod (Region 30, 263 210 km2; 520k to 665k to 740k). Finally, across Northern Asia, the eight Siberian functional zones (Regions 31-38) comprised the Pomor-Mangazeya maritime and Taz-Turukhansk Arctic route (Region 31, 1 760 905 km2; 16k to 26k to 18k), the Tobolsk-Irtysh-Middle Ob water highway (Region 32, 871 344 km2; 42k to 88k to 158k), the southern Tara-Tomsk-Kuznetsk fortified agricultural frontier (Region 33, 359 192 km2; 38k to 86k to 254k), the nomadic Baraba-Altai-Minusinsk-Tuva steppe borderlands (Region 34, 821 185 km2; 74k to 84k to 100k), the Ket-Yeniseysk-Angara-Ilimsk-Irkutsk portage spine (Region 35, 1 931 504 km2; 28k to 44k to 98k), Transbaikal Dauria and the Upper Amur frontier (Region 36, 1 753 934 km2; 34k to 40k to 62k), the central Lena-Vilyuy-Aldan Yakutsk basin (Region 37, 1 373 412 km2; 38k to 42k to 47k), and the Northeast Arctic and Pacific *yasak* frontier (Region 38, 3 924 538 km2; 30k to 25k to 23k).

Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
1. Terra Mariana / Swedish Estland & Livland	260	330	390	270	290	260	360	340	390	440	300
2. Courland, Semigallia & Polish Inflanty	140	175	210	190	210	200	260	210	240	280	200
3. GDL Core: Vilnius, Trakai & Samogitia	460	610	720	780	850	890	1 060	640	710	810	650
4. Vyborg Fief & Kexholm / Old Finland	75	92	105	88	95	80	110	105	118	130	110

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Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
5. Polotsk & Vitebsk Voivode-ships (GDL)	240	320	390	350	410	440	560	270	310	360	370
6. Grodno & Novo-grudok Voivode-ships (GDL)	280	385	470	520	580	620	740	420	470	520	490
7. Minsk, Orsha & Mogilev Dnieper (GDL)	260	350	430	460	510	550	680	330	380	430	430
8. Brest-Litovsk & Pinsk-Turov Polesia (GDL)	170	245	310	340	380	410	520	260	290	320	260
9. Ruthenian/ Galicia, Belz & Carpathian Rus	820	1 020	1 180	1 290	1 420	1 510	1 750	1 380	1 450	1 580	1 620
10. Volhynian Voivode-ship (Lutsk & Ostroh)	420	530	620	680	750	790	920	580	610	660	680
11. Podolian & Bratslav Voivode-ships	350	450	540	610	690	740	910	490	380	440	510
12. Right-Bank Kyiv Regiments (Chyhyryn)	280	360	440	520	620	690	920	520	360	320	340

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Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
13. N. Left-Bank Hetmanate (Chernihiv-Nizhyn)	210	260	300	330	370	390	490	560	620	720	790
14. S. Left-Bank Hetmanate (Poltava-Lubny)	110	145	180	220	270	320	460	540	610	710	760
15. Sloboda Ukraine Regiments (Kharkiv-Sumy)	50	65	80	95	110	120	160	250	310	380	400
16. Zaporozhian Sich & Pontic Wild Fields	60	70	80	90	100	105	130	135	145	165	180
17. Crimean Khanate & Ottoman Kefe	210	260	290	310	330	335	360	365	375	385	370
18. Moscow & Zamoskovny Core (Vladimir-Tver)	1 950	2 620	3 050	2 780	2 950	2 280	2 850	3 080	3 320	3 620	3 720
19. Novgorod & Pskov Five Pyatiny	920	1 170	1 340	960	1 040	710	930	1 010	1 090	1 180	1 230
20. Olonets Karelia & Baltic Ingria (St P.)	200	250	280	220	240	180	220	230	250	270	350

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Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
21. Smolensk & Seversk Border-lands (Bryansk)	480	600	680	640	690	510	640	580	640	710	730
22. Pomorye, Vologda, Dvina & Pechora North	210	280	340	380	410	420	460	470	480	450	420
23. Ryazan, Tula, Oryol & Great Oka Abatis	680	850	980	940	1 020	820	1 080	1 180	1 290	1 420	1 520
24. Belgorod & Sevsk Razryads (Voronezh-Kursk)	190	240	290	340	410	380	620	820	980	1 220	1 440
25. Kazan, Nizhny Novgorod & Simbirsk Line	780	890	940	960	1 020	980	1 180	1 320	1 460	1 620	1 760
26. Vyatka, Great Perm & Stroganov Urals	220	280	330	360	390	405	480	520	560	610	680
27. Bashkir Four Darugas (Ufa) & Yaik Frontier	180	200	215	225	235	240	260	275	290	310	340
28. Lower Volga, Astrakhan & Kalmyk Steppe	240	260	270	265	275	280	320	350	375	400	430

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Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
29. Don Cossack Host (Cherkassk) & Azov	90	100	110	120	135	130	165	190	215	240	250
30. North Caucasus (Circassia, Kabarda, Dagestan)	520	565	595	610	625	635	665	680	695	715	740
31. Pomor-Mangazeya Maritime & Taz Portage	16	17	18	19	24	28	26	22	19	18	18
32. Tobolsk-Irtysh-Middle Ob Portage Highway	42	45	47	49	56	64	88	105	118	138	158
33. S. Tobol-Tara-Tomsk-Kuznetsk Ploughland	38	42	45	47	52	60	86	118	145	195	254
34. Nomadic Baraba-Altai-Minusinsk Steppe	74	77	80	81	82	82	84	82	80	88	100
35. Ket-Yeniseysk-Angara-Ilimsk-Irkutsk Spine	28	30	31	32	33	34	44	56	66	82	98
36. Trans-baikal Dauria & Upper Amur (Nerchinsk)	34	36	37	38	39	39	40	44	48	55	62

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Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
37. Central Lena-Vilyuy-Aldan (Yakutsk Alaas)	38	40	41	42	42	41	42	43	44	45	47
38. NE Arctic & Pacific Yasak (Kolyma-Kamchatka)	30	30	31	30	29	27	25	25	25	24	23
Baltic, Lithuania & Old Finland (Reg. 1-4)	935	1 207	1 425	1 328	1 445	1 430	1 790	1 295	1 458	1 660	1 260
Belarus GDL Subtotal (Reg. 5-8)	950	1 300	1 600	1 670	1 880	2 020	2 500	1 280	1 450	1 630	1 550
Ukraine & Crimea Subtotal (Reg. 9-17)	2 510	3 160	3 710	4 145	4 660	5 000	6 100	4 820	4 860	5 360	5 650
Russia European & Steppe (Reg. 18-30)	6 660	8 305	9 420	8 800	9 440	7 970	9 870	10 705	11 645	12 765	13 610
Russia Siberian Zones (Reg. 31-38)	300	317	330	338	357	375	435	495	545	645	760
Russia Subtotal (Reg. 18-38)	6 960	8 622	9 750	9 138	9 797	8 345	10 305	11 200	12 190	13 410	14 370

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Historical Region / Subtotal	1400AD	1500AD	1550AD	1580AD	1600AD	1615AD	1648AD	1667AD	1678AD	1700AD	1720AD
Total Domain (All 38 Period Regions)	11 355	14 289	16 485	16 281	17 782	16 795	20 695	18 595	19 958	22 060	22 830

Table 1. Regional population estimates and polity subtotals across all thirty-eight historical regions in thousands (1400AD - 1720AD).

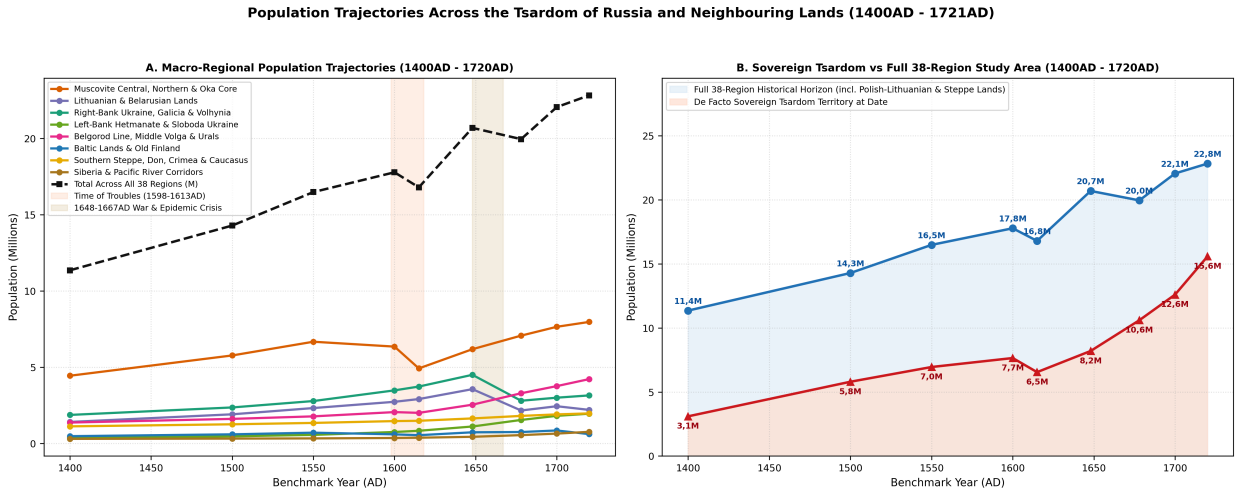


Figure 2. Regional and macro-regional population trajectories across Russia, Ukraine, Belarus, Lithuania, and the Baltic provinces (1400AD - 1720AD).

Fiscal Cadastres, Ploughland Audits, Household Censuses, and Revisions

Demographic reconstruction across the thirty-eight regions between 1400AD and 1721AD rests on six complementary fiscal and ecclesiastical accounting traditions that progressively replaced indirect arable-unit assessments with household, hearth, and individual soul enumerations [3][7][21][22]. In the Muscovite core and northern provinces (Regions 18-26), late 15th- and 16th-century registration relied on the *pistsovye knigi* (scribe land cadastres), which assessed obligations in units of arable land known as the *sokha* (400 to 800 *chetverti* of good arable land in three fields, scaled by service, church, or state-peasant tenure) and the Novgorodian *obzha* (Regions 19-20) [2][23]. Because peasants reacted to wartime fiscal pressure by abandoning assessed ploughland or merging nuclear families into extended households, cadastral *dvor* counts require calibration against empirically observed household multipliers of 6,0 to 8,5 persons [1][19].

The shift from land-area (*pozemelnoe*) to household (*podvornoe*) taxation in the Tsardom of Russia produced the comprehensive enumerations of 1646AD and 1678AD (*perepisnye knigi*) [7][21]. Cross-checking the 1678AD household books against the 1710AD wartime audit and Peter I's First Poll-Tax Revision (1719-1721AD, verified by military *popolnenie* commissions in 1721-1724AD), Ya. E. Vodarsky

(1977) and V. M. Kabuzan (1963) demonstrated that pre-Petrine household registers omitted rural clergy, *sluzhilye lyudi* servicemen, and untaxed *bobyli* or *zadvornye* dependents while encouraging artificial household amalgamation to 8,5-9,2 persons (4,2-4,6 males) per *dvor* [7][8]. Replacing the household levy with the individual male capitation tax (*podushnaya podat*) in 1718-1719AD removed the fiscal incentive to conceal split families and registered 7,18M male souls (14 370k total inhabitants of both sexes across Regions 18-38 in 1720AD) [8][24].

Along the western Baltic and Fennoscandian littoral (Regions 1, 2, and 4), demographic series are anchored by Livonian and Swedish *Hakenrevisionen* (ploughland *adramaa* and *haken* audits of 1586-1591AD, 1601AD, 1638AD, 1688AD, and 1712AD), Lutheran parish registers (*kirikuraamatud* introduced across Estland, Livland, and Vyborg-Kexholm from the mid-17th century), Courlandian ducal domain inventories, and Swedish poll-tax (*mantalslangder*) rolls reconstructed by H. Palli (1996), S. Vahtre (1973), E. Dunsdorfs (1962), and V. Saloheimo (1976) [3][10][11][25]. In core Lithuania (Region 3), Belarus (Regions 5-8), and Polish Crown Ukraine (Regions 9-12), estimates rest on the 1528AD Grand Duchy of Lithuania Military Census (*Popis voyska*, recording 19 842 noble cavalrymen at a statutory ratio of 1 horseman per 8 peasant *sluzhby* alongside grand-ducal and church estates), the 1557AD *Voloka* agrarian cadastres (*Pomiara wloczna*), Crown royal *lustracje* (1564-1565AD, 1616AD, 1629AD), and *podymne* hearth-tax registers of 1629AD, 1648-1650AD, 1667AD, 1690AD, and 1717AD [4][5][9][13][26]. Finally, the Cossack Hetmanate and Sloboda Ukraine (Regions 12-15), Crimea (Region 17), and the eight Siberian zones (Regions 31-38) draw on the 1649AD Zboriv Register, the 1666AD Left-Bank *perepysni knyhy*, Ottoman *tahrir defterleri* of Kefe, and Siberian *yasachnye* and *dozornye knigi* [6][20][27][28].

Documentary Corpus	Period	Primary Accounting Unit	Demographic Multiplier	Regional Coverage	Principal Scholarship
Baltic Hakenre-visionen & Mantalslangder	1561-1715AD	Adramaa / Haken, farmstead (talu) & parish soul	7,5 - 9,5 persons / talu farmstead	Regions 1, 2 & 4 (Estland, Livland, Courland, Vyborg)	Palli 1996; Vahtre 1973; Dunsdorfs 1962; Saloheimo 1976
Lithuanian 1528AD Popis & Voloka Books	1528-1600AD	Cavalry horse, sluzhba & voloka (21,36 ha)	6,5 - 7,5 persons / voloka-sluzhba	Regions 3, 5-8 & 10 (Lithuania, Belarus & Volhynia)	Lowmianski 1983; Ochmanski 1982; Vasiliauskas 2012
Polish Crown Lustracje & Podymne Hearths	1564-1717AD	Taxable hearth (dym) & lan arable unit	6,0 - 6,5 persons / rural dym	Regions 2-3 & 5-12 (Inflanty, GDL & Crown Rus)	Morzy 1965; Karpachev & Kozlovsky 1972; Yakovenko 2006
Novgorod & Moscow Pistsovye Knigi	1495-1594AD	Sokha, obzha & peasant household (dvor)	5,5 - 7,0 persons / dvor	Regions 18-23 (Zamoskovny, Pyatiny, Ingria, Oka)	Kopanev 1959; Shapiro 1971, 1974; Selin 2020
Zboriv Register, 1666AD Books & Tahrir	1520-1700AD	Cossack kurin/ dvor, posad dvor & Ottoman hane	5,5 - 7,0 persons / household	Regions 12-17 (Hetmanate, Sloboda & Crimea)	Krypiakevych 1954; Irtenina 2022; Ozturk 2019

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Documentary Corpus	Period	Primary Accounting Unit	Demographic Multiplier	Regional Coverage	Principal Scholarship
1646AD & 1678AD Perepisnye Knigi	1646AD; 1678AD	Taxable household (tyaglyy dvor)	8,0 - 9,2 persons / enlarged dvor	Regions 18-29 (Muscovite Core & Frontier Razryads)	Vodarsky 1973, 1977; Avdeev 2025
Siberian Yasachnye & Dozornye Knigi	1595-1719AD	Yasak-paying adult male (18-50 years)	4,0 - 4,2 total pop / yasak payer	Regions 31-38 (8 Siberian Portage & Yasak Zones)	Dolgikh 1960; Pokrovsky 1984; Ushnitskiy 2017b
First Poll-Tax Revision (Revizskie Skazki)	1719-1724AD	Individual male soul (revizskaya dusha)	2,00 - 2,02 total pop / male soul	Regions 1, 4, 13-15 & 18-38 (Petrine Empire)	Kabuzan 1963, 1990; Ulyanova 2016; Mironov 2007

Table 2. Principal fiscal cadastres, ploughland audits, parish registers, and household censuses used for regional demographic reconstruction (1400AD - 1721AD).

The Baltic Littoral, Courland-Inflanty, Core Lithuania, and Old Finland (Regions 1-4)

Terra Mariana to Swedish Estland and Livland (Region 1) and Courland-Inflanty (Region 2)

Across the Eastern Baltic littoral (Regions 1 and 2, covering 113 513 km2 combined), late medieval political authority belonged to the ecclesiastical and knightly confederation of *Terra Mariana* (the Archbishopric of Riga, the Bishoprics of Dorpat, Osel-Wiek, and Courland, and the Livonian Branch of the Teutonic Order) alongside Hanseatic mercantile municipalities led by Riga (8k in 1400AD -> 12k in 1550AD), Reval / Tallinn (5,5k -> 8k), Dorpat / Tartu (4,5k -> 6,5k), Pernau / Parnu (1,6k -> 2,6k), Narva (1,8k -> 3,2k), Wenden / Cesis (1,5k -> 2,4k), and Arensburg / Kuressaare [3][10][29]. Favoured by a century of relative internal peace after 1435AD and rising rye, flax, and timber exports to Western Europe, Terra Mariana / Estland and Livland (Region 1, 76 229 km2) grew from 260k in 1400AD to 330k in 1500AD and 390k in 1550AD, while southern Courland, Semigallia, and Latgale (Region 2, 37 284 km2) rose from 140k to 210k [3][10][25].

Ivan IV's invasion of Narva and Dorpat in 1558AD shattered the medieval confederation and initiated seventy years of intermittent Baltic warfare. During the Livonian War (1558-1583AD) and the subsequent Polish-Swedish wars of 1600-1629AD - compounded by the extreme volcanic-winter crop failure, famine, and plague of 1601-1603AD - rural *adramaa* farmstead occupancy across mainland Estland and northern Livland dropped by more than half [3][14]. Region 1 fell by 30,8% between 1550AD (390k) and 1580AD (270k) and stood at 260k in 1615AD before the Truce of Altmark (1629AD) and Treaty of Bromsebro (1645AD, ceding Osel/Saaremaa from Denmark) consolidated Swedish Estland and Livland [3][30]. Under 17th-century Swedish administration, inward rural migration of Finnish, Russian, Latvian, and Lithuanian settlers, parish schooling, and Charles XI's *Reduktion* of noble estates in the 1680sAD drove vigorous demographic recovery: Region 1 climbed to 360k in 1648AD, 390k in 1678AD, and 440k at the turn of the century (c. 1695-1700AD), despite the severe Great Famine of 1695-1697AD (*Suur nalg*), when cold,

rain-soaked summers destroyed two consecutive harvests and killed roughly 70k-75k inhabitants across Estland and Livland while Swedish authorities continued shipping Baltic manor grain to Stockholm and Finland [3][14][31].

South of the Daugava (Western Dvina), the Pact of Vilnius (1561AD) secularised the southern Ordensstaat into the hereditary Duchy of Courland and Semigallia under Gotthard Kettler as a fief of the Polish-Lithuanian Commonwealth, while eastern Latgale remained under direct Commonwealth sovereignty after 1629AD as the Voivodeship of Inflanty (Dunaburg / Daugavpils, Rezhitsa / Rezekne, and Ludza) [10][32]. Shielded from the worst fighting of the Livonian War (dipping mildly from 210k in 1550AD to 190k in 1580AD), Region 2 expanded to 260k in 1648AD under the mercantilist manufactures, shipyards, and Atlantic-Baltic trade of Duke Jacob Kettler (1642-1682AD) at his capital Mitau / Jelgava (5,5k in 1648AD), the shipbuilding port of Windau / Ventspils (3,2k), Libau / Liepaja (3,5k), Goldingen / Kuldiga (3,0k), and Bauske (2,5k) [10][33]. After suffering a 19,2% contraction (260k -> 210k) during the 1658-1660AD Swedish occupation of Courland and Muscovite capture of Dunaburg (briefly renamed *Borisoglebsk* in 1656-1666AD), Region 2 recovered to 280k by 1700AD [10][34].

Both Baltic regions then experienced an unprecedented demographic catastrophe during the Great Northern War (1700-1710AD) and the Great Baltic Plague of 1710-1711AD [3][15][33][34]:

Region 1 (Estland and Livland) plunged by 43,2% in a single decade, from 440k in 1700AD to 250k in 1710AD, as Russian and Swedish campaigns around Narva (1700AD, 1704AD), Dorpat (besieged in 1704AD and systematically demolished in 1708AD with its burghers deported to Vologda), Pernau, and Riga culminated in the *Yersinia pestis* epidemic of summer-autumn 1710AD, which killed over half the besieged population of Riga (15k to 6,5k), roughly two-thirds of Reval (5k to 4,5k), and up to 70% of western Estland coastal parishes before the Baltic knighthoods signed the Capitulations of 1710AD (confirmed by the Treaty of Nystad in 1721AD, with population recovering to 300k in 1720AD) [3][15][30].

Region 2 (Courland, Semigallia, and Polish Inflanty) fell by 39,3%, from 280k in 1700AD to 170k in 1710AD (recovering to 200k by 1720AD), as Saxon, Swedish, and Russian armies fought across the Daugava (1701AD) and Gemauerthof (1705AD) and the 1710AD plague wiped out nearly half of Courland's rural inhabitants and reduced Mitau from 6,8k to 2,6k [10][33][34].

Grand Duchy of Lithuania Core: Vilnius, Trakai, and Samogitia (Region 3)

Immediately south of Courland and Livonia lay the political and ethnic Lithuanian core of the Grand Duchy of Lithuania: Region 3 (65 877 km2), uniting the western districts of the Vilnius and Trakai Voivodeships with the Eldership of Samogitia (*Zemaitija*) [4][35]. Untouched by Tatar steppe raiding north of the Neman (Nemunas) and stimulated by German/Prussian and Baltic river commerce through the Hanseatic factory at Kaunas, Region 3 grew steadily from 460k in 1400AD to 610k in 1500AD and 720k in 1550AD. In Henryk Lowmianski's and Jerzy Ochmanski's reconstructions of the 1528AD GDL Military Census (*Popis voyska*), ethnic Lithuania and Samogitia exhibited the highest rural settlement densities in the entire Grand Duchy [4][26]. Following Sigismund II Augustus's 1557AD Volok Reform (*Valaku reforma*), which reorganised grand-ducal and magnate domains (such as the vast Siauliai Crown Economie) into regular three-field *voloka* villages and chartered scores of market towns (*miesteliai*), Region 3 expanded to 850k in 1600AD and 1 060k in 1648AD [35][36]. At its mid-17th-century zenith, the Grand-Ducal capital Vilnius (seat of the 1579AD Jesuit University, the Lithuanian Tribunal, and thriving Ruthenian, Polish, Lithuanian, Jewish, German, and Karaite communities) reached ~28k inhabitants, flanked by Kaunas (~15k), the Radziwill Protestant commercial hub of Kedainiai (~6,5k), Trakai (~5k), Siauliai (~3,5k), Raseiniai (~3,2k, parliamentary seat of Samogitia), and Telsiai (~2,6k) [35][36][37].

This Golden Age ended abruptly in August 1655AD when Tsar Aleksey Mikhailovich's armies and Ivan Zolotarenko's Cossacks captured Vilnius - the first foreign capture of the Lithuanian capital in its history -

sparking a seventeen-day conflagration, while Swedish forces occupied Samogitia and Kedainiai [13][35]. Six years of Muscovite occupation in Vilnius and Trakai (1655-1661AD), famine, the 1657-1658AD plague, and artisan deportations reduced Region 3 by 39,6%, from 1 060k in 1648AD to 640k in 1667AD (with Vilnius shrinking to 11,5k and Kaunas to 6,0k) [4][13]. Although post-war reconstruction lifted Region 3 back to 710k in 1678AD and 810k in 1700AD, the Great Northern War (during which Swedish, Russian, Saxon, and rival Sapieha-Oginski Lithuanian confederate armies ravaged Samogitia and Vilnius) and the devastating Great Plague of 1708-1711AD killed more than a quarter of the population, driving Region 3 down to 580k in 1710AD (-28,4%) before an initial recovery to 650k in 1720AD [16][32][35].

Vyborg Fief, Kexholm County, and the Origins of Old Finland (Region 4)

On the north-western Karelian Isthmus, Lake Saimaa, and the western shore of Lake Ladoga, Region 4 (49 685 km2) bridges the Swedish-Finnish and Novgorodian-Russian worlds [11][38]. Throughout the 15th and early 16th centuries, the western half of this territory formed the Swedish frontier province of Vyborg Fief (*Viipurin linnalaani*), guarded by the stone castle of Vyborg / Viipuri (chartered 1403AD, 2,2k in 1400AD -> 3,6k in 1550AD) and the inland Savo lake fortress of Nyslott / Savonlinna (*Olavinlinna*, founded 1475AD), while the eastern Ladoga shore belonged to Novgorod's and Muscovy's Korela (*Kexholm / Priozersk*) uezd [23][38]. Combined population grew from 75k in 1400AD to 105k in 1550AD before the 25-year Russo-Swedish "Long Wrath" (*Pitka viha*, 1570-1595AD) and Swedish occupation of Kexholm (1580-1595AD) depressed regional numbers to 88k in 1580AD and 80k during the *Smuta* / Ingrian War in 1615AD [38][39].

When the Treaty of Stolbovo (February 1617AD) transferred Kexholm County (*Kakisalmen laani*) to the Swedish Empire, Region 4 underwent a near-total ethno-confessional replacement documented in Swedish tax rolls and Muscovite *pistsovye knigi* by V. Saloheimo (1976), K. Katajala (2017), O. M. Fishman (2020), and Yu. V. Stepanova (2024) [11][38][40][41]. Facing heavy Swedish crown rents, noble fief (*forlaning*) exactions, and Lutheran proselytising under Bishop Isaac Rothovius's successors, Orthodox Karelian families migrated *en masse* across the border into Russia - first during the 1620s-1640sAD (some 25k migrants) and decisively during the Russo-Swedish War of 1656-1658AD (*Ruptuuri*), when an additional 20k-25k Orthodox Karelians withdrew with Russian troops and were settled by Tsar Aleksey Mikhailovich on crown lands in Bezhetskiy Verkh, Tver, Torzhok, Vyshny Volochyok, and Valdai (Regions 18-19), creating the enduring Tver Karelian population [40][41]. Simultaneously, Lutheran peasants from Savonia (*Savo*) and Vyborg Karelia poured into the vacated Kexholm parishes to clear slash-and-burn (*kaski*) rye plots, lifting Region 4 to 110k in 1648AD and 130k by 1700AD alongside newly chartered Saimaa-Gulf of Finland tar-export towns at Villmanstrand / Lappeenranta (1649AD) and Vehkalahti / Fredrikshamn (Hamina) (1653AD) [11][38]. The Finnish Great Famine of 1696-1697AD (*Suuret kuolonvuodet*), followed by Peter I's 1710AD Sieges of Vyborg and Kexholm (conceived as the northern security glacis for Saint Petersburg) and the Great Wrath occupation (*Isoviha*), cut Region 4 by 26,9% to 95k in 1710AD before recovering to 110k in 1720AD upon formal Russian annexation of Vyborg and Kexholm ("Old Finland", *Vanha Suomi*) at the Treaty of Nystad (1721AD) [11][38].

Western / Baltic Region	Area	Principal Urban & Fortress Centres	1648AD / 1700AD	Crisis Trough (% Drop)	1720AD
1. Estland & Livland (Terra Mariana / Ostsee)	76 229 km2	Riga, Reval (Tallinn), Dorpat (Tartu), Narva, Pernau, Wenden, Arensburg	360k / 440k	250k (1710AD, -43,2%)	300k
2. Courland, Semigallia & Polish Inflanty	37 284 km2	Mitau (Jelgava), Libau, Windau, Goldingen, Bauske, Dunaburg, Rezhitsa	260k / 280k	170k (1710AD, -39,3%)	200k

(Continued from previous page)

Western / Baltic Region	Area	Principal Urban & Fortress Centres	1648AD / 1700AD	Crisis Trough (% Drop)	1720AD
3. GDL Core: Vilnius, Trakai & Samogitia	65 877 km2	Vilnius, Kaunas, Trakai, Kedainiai, Siauliai, Raseiniai, Telsiai	1 060k / 810k	640k (1667AD) / 580k (1710AD)	650k
4. Vyborg Fief & Kexholm / Old Finland	49 685 km2	Vyborg (Viipuri), Kexholm (Priozersk), Nyslott, Fredrikshamn, Villmanstrand	110k / 130k	80k (1615AD) / 95k (1710AD)	110k
Western Rim Subtotal (Regions 1-4)	229 075 km2	27 chartered Hanseatic, Magdeburg, ducal, and Swedish staple towns	1 790k / 1 660k	1 095k (1710AD, -34,0% vs 1700AD)	1 260k

Table 3. Population summary of the Baltic provinces, Lithuania, and Old Finland (1400AD - 1720AD), showing pre-war peaks, epidemic troughs, and 1720AD totals.

Muscovite Core, Novgorod Pyatiny, Karelia-Ingria, and Pomorye (Regions 18-22)

Consolidation of the Forest Core and Separation of Novgorod Pyatiny (Region 19) from Karelia-Ingria (Region 20)

At the opening of the 15th century, the five northern and western forest regions of Russia (Regions 18-22, totalling 3 760k in 1400AD) were recovering from recurrent waves of the Black Death (1352-1353AD, 1364-1366AD, 1417AD, and 1425-1427AD) and the destructive campaigns of Tokhtamysh (1382AD) and Edigu (1408AD) [1][42]. Within Moscow and Zamoskovny Core (Region 18, 277 154 km2; 1 950k in 1400AD -> 3 050k in 1550AD), internal agricultural colonisation advanced along the watershed ridges (*vodorazdely*) of the Moscow-Klyazma basin, the fertile Suzdal *Opolye*, Tver, Yaroslavl, Kostroma, and Kaluga, led by peasant clearance and monastic settlement networks around the Trinity-Sergius Lavra [2][23].

Under Ivan III (1462-1505AD) and Vasily III (1505-1533AD), Moscow annexed the Novgorod Republic (1478AD), the Grand Principality of Tver (1485AD), Vyatka (1489AD), the Upper Oka and Seversk principalities (1494-1503AD), Pskov (1510AD), Smolensk (1514AD), and Ryazan (1521AD) [43][44][45]. Because the Ilmen-Pskov arable heartland and the Finno-Karelian / Neva borderlands followed distinct institutional and military trajectories between 1478AD and 1721AD, our 38-region partition separates Region 19 from Region 20 (while western Vyborg and Ladoga Kexholm are tracked separately in Region 4):

Region 19: Novgorod and Pskov Lands (Five Pyatiny and Velikie Luki) (159 354 km2; 920k in 1400AD to 1 170k in 1500AD to 1 340k in 1550AD): Encompassing the core Shelonskaya, Derevskaya, Bezhetskaya, and southern Vodskaya/Obonezhskaya *pyatiny* around Lake Ilmen, the Volkhov, and the Lovat, together with the Pskov Republic (Pskov, Izborsk, Ostrov, Opochka, Gdov) and the southern border fortress of Velikie Luki [23][44]. Following the confiscation of Novgorodian boyar and archiepiscopal estates in 1478-1489AD and the settlement of Moscow cavalry *pomeshchiki*, the surviving *pistsovye knigi* of 1495-1505AD and the 1540s-1550sAD document a 14,5% demographic expansion between 1500AD and 1550AD, with Veliky Novgorod (~33k-35k in 1500-1550AD) and Pskov (~28k-30k inside the Okolny Gorod wall) ranking as the second and third cities of the Tsardom [12][23].

Region 20: Olonets Karelia and Baltic Ingria (Ivangorod, Oreshek, and St Petersburg 1703AD) (205 031 km2; 200k in 1400AD to 280k in 1550AD to 180k in 1615AD to 350k in 1720AD): Spanning the Neva-Gulf of Finland littoral (*Izhora / Ingria*: Ivangorod founded 1492AD opposite Narva, Yam, Koporye, and Oreshek / Noteburg at the Neva outlet) together with Olonets-Onega Karelia (*Zaonezhskie* and *Lopatinskie pogosty*, Olonets fortress founded 1649AD) [23][39]. Unlike the Ilmen-Pskov grain belt (Region 19), Region 20 combined slash-and-burn (*podseka*) agriculture, Ladoga-Onega fisheries, bog-iron smelting (*domnitsy*, and later Petrine Olonets ironworks at Petrovskaya Sloboda in 1703AD), and strategic Neva-Ingrian fortresses ceded to Sweden under the Treaty of Stolbovo (1617AD) before Peter I's reconquest and foundation of Saint Petersburg (1703AD) [12][39].

Region 21: Smolensk and Seversk Borderlands (Smolensk, Bryansk, Roslavl, and Upper Desna) (83 068 km2; 480k in 1400AD to 680k in 1550AD to 730k in 1720AD): Guarded the strategic Upper Dnieper and Desna-Bryansk corridor between Moscow and the Grand Duchy of Lithuania, anchored by Fyodor Kon's brick citadel at Smolensk (1595-1602AD), Dorogobuzh, Vyazma, Roslavl, and Bryansk [1]. Region 22: Pomorye, Vologda, Northern Dvina (Arkhangelsk), and Vychegda-Pechora Komi Lands (1 385 055 km2; 210k in 1400AD to 340k in 1550AD to 480k in 1678AD to 420k in 1720AD): The vast *chernososhnye* (state-peasant) maritime, salt-boiling, fur-trading, and Zyryan Komi region of the Russian North. After English navigator Richard Chancellor reached the Northern Dvina in 1553AD, the White Sea trade route stimulated rapid urban growth along the Sukhona-Dvina axis linking Vologda (13k in 1550AD), Veliky Ustyug (5k), Solvychegodsk (4k), Kholmogory (8k), Arkhangelsk (founded 1584AD), Kargopol, Kola, and the Pechora portage outpost of Pustozersk (1499AD) [1][42]. Region 22 peaked at 480k in 1678AD before declining to 420k by 1720AD as Petrine military conscription, naval timber levies, and the 1713-1718AD decrees diverting foreign maritime trade from Arkhangelsk to Saint Petersburg (Region 20) drew population out of the White Sea basin [7].

The Oprichnina, the Livonian War, and the Pustoshi Crisis of the 1570s-1580sAD

The demographic expansion of the old Muscovite heartland was abruptly reversed during the second half of the reign of Ivan IV (1547-1584AD). Four overlapping shocks struck Regions 18, 19, 20, and 21 between 1565AD and 1583AD [12][39][46]:

Fiscal Over-Taxation and the Livonian War (1558-1583AD): Twenty-five years of continuous mobilisation and extraordinary wartime levies (*pyatinnye dengi*, *yam* post obligations, and siege-artillery transport) exhausted the tax-paying peasantry (*tyaglye krestyane*) and *pomeshchik* service estates across the Novgorod, Pskov, Ingrian, Smolensk, and Zamoskovny districts [12][39]. The Oprichnina Terror (1565-1572AD): Ivan IV's division of the realm into the *Zemshchina* and the *Oprichnina* entailed forced estate confiscations, predatory exactions, and the punitive expedition against Tver, Torzhok, Klin, and Veliky Novgorod in January-February 1570AD, killing thousands of inhabitants and paralysing regional grain trade [1][12].

Subsistence Famine and Plague (1569-1571AD): Early frosts and harvest failure in 1569-1570AD triggered severe famine across central and north-western Russia, compounded by a lethal plague epidemic (*morovoe povetrie*) throughout Regions 18-20 [12]. Crimean Tatar Devastation (1571-1572AD): In May 1571AD, Khan Devlet I Giray bypassed the Oka defences (Region 23) and burned the wooden suburbs of Moscow (Region 18), killing tens of thousands and carrying off an estimated 60k-100k captives before Prince Mikhail Vorotynsky defeated the Crimean host at the Battle of Molodi in 1572AD [47][48].

The cadastral books compiled in the 1580sAD and early 1590sAD document widespread rural abandonment (*zapustenie*) [1][12]. In Novgorod and Pskov (Region 19), between 75% and 90% of previously cultivated *obzhi* were registered as deserted wasteland (*pustoshi* or *porozhnie zemli*); regional population dropped by 28,4%, from 1 340k in 1550AD to 960k in 1580AD, while Olonets Karelia and Baltic Ingria (Region 20) fell by 21,4% (280k -> 220k) during Pontus De la Gardie's Swedish capture of Ivangorod,

Yam, and Koporye (1581-1590AD, recovered by Russia at the 1595AD Truce of Teusina) [12][39]. Moscow and Zamoskovny Core (Region 18) contracted from 3 050k (1550AD) to 2 780k (1580AD), and Smolensk and Seversk (Region 21) declined from 680k to 640k [1][45]. Conversely, fugitive peasants moved north into Pomorye (Region 22, 340k -> 380k), south into the Belgorod and Sevsk Razryads (Region 24, 290k -> 340k), and east into Kazan and the Middle Volga (Region 25, 940k -> 960k) and Perm-Urals (Region 26, 330k -> 360k), prompting the Muscovite crown to introduce the *zapovednye leta* ("forbidden years", from 1581AD) suspending peasant departure on St George's Day (*Yuryev den*) [19][46].

Belarusian GDL Voivodeships and Polish Crown Ukraine (Regions 5-14: 1400AD - 1648AD)

The Four Belarusian GDL Voivodeships (Regions 5-8) and the Voloka Agrarian Reform

In sharp contrast to late 16th-century Muscovy, the four Belarusian regional clusters of the Grand Duchy of Lithuania (Regions 5-8, covering 246 646 km2) experienced sustained demographic and urban expansion between 1400AD and 1648AD, with combined Belarusian population rising from 950k (1400AD) to 1 300k (1500AD), 1 600k (1550AD), 1 880k (1600AD), and 2 500k on the eve of the mid-17th-century wars (1648AD) [9][13][26]:

Region 5: Polotsk and Vitebsk Voivodeships (32 576 km2): Anchored by the Western Dvina export artery to Riga and the commercial cities of Polotsk (~20k in 1648AD), Vitebsk (~17k), Drissa, and Lepel, population expanded from 240k (1400AD) to 390k (1550AD), dipped temporarily to 350k (1580AD) during Ivan IV's occupation of Polotsk (1563-1579AD) and Stephen Bathory's reconquest, and climbed to 560k in 1648AD [13][49]. Region 6: Grodno and Novogrudok Voivodeships (48 968 km2): Comprising Black Ruthenia (*Chornaya Rus*) and the fertile Ponemanje basin (Grodno ~13,5k, Novogrudok ~9k, Slonim ~7,2k, Lida ~6k, and Volkovysk ~5,5k), this densely cultivated region grew from 280k (1400AD) to 470k (1550AD) and 740k in 1648AD [13][26].

Region 7: Minsk Voivodeship and Orsha-Mogilev Dnieper Corridor (102 278 km2): Spanning the central Belarusian watershed and the Upper Dnieper-Berezina-Sozh river arteries, population rose from 260k (1400AD) to 430k (1550AD) and 680k (1648AD), led by Mogilev (~25k in 1648AD, the largest city in Belarus and premier transit market between Moscow and Poland-Lithuania), Minsk (~13k), Slutsk (~12,5k), Orsha (~9,5k), Mstislavl (~9k), the Radziwill residence at Nesvizh (~7,5k), Gomel, and Bobruysk [9][49].

Region 8: Brest-Litovsk Voivodeship and Pinsk-Turov-Mozyr Polesia (62 824 km2): Encompassing the Western Bug gateway at Brest-Litovsk (~15k in 1648AD, granted Magdeburg Law in 1390AD and site of the 1596AD Church Union) alongside the Pripyat river-marshland towns of Pinsk (~12k), Mozyr (~6k), and Turov, population tripled from 170k (1400AD) to 310k (1550AD) and 520k in 1648AD [9][13].

This sustained expansion rested on Queen Bona Sforza's and Sigismund II Augustus's Voloka Reform (*Ustava na voloki*, 1557AD), which consolidated scattered peasant *dvorishcha* into regular three-field linear villages divided into standard 21,36 ha *voloka* plots, paired with the chartering of more than 60 Magdeburg-law municipalities and over 300 noble market towns (*miasteczka*) [13][49].

Crown Ukraine, Right-Bank Kyiv, and the Left-Bank Colonisation Boom (Regions 9-14: 1400AD - 1648AD)

Across the Ukrainian and Crimean territories (Regions 9-17, covering 607 825 km2), total population rose from 2 510k in 1400AD to 3 160k (1500AD), 3 710k (1550AD), 4 660k (1600AD), and 6 100k in 1648AD [5][27]. Following the Union of Lublin (1569AD), which transferred Volhynia, Podolia-Bratslav, and the Kyiv

Voivodeship from Lithuania to the Polish Crown, each Ukrainian region followed a distinct settlement trajectory:

Region 9: Ruthenian and Belz Voivodeships (Galicia), Bukovina, and Hungarian Carpathian Rus (71 558 km2): The most densely settled East Slavic territory under Polish Crown rule (paired with Moldavian Bukovina and Hungarian Transcarpathia), expanding from 820k (1400AD) to 1 180k (1550AD) and 1 750k in 1648AD (5 persons/km2). Its metropolitan capital Lviv (~25k in 1648AD) linked Black Sea and Levantine trade with Central Europe, flanked by Drohobych (~7,2k), Brody (~6,5k), Halych, Ternopil, and Zhovkva [5][27].

Region 10: Volhynian Voivodeship (45 176 km2): Seat of the great Ruthenian princely houses (Ostrogski, Zaslawski, Zbaraski, Sanguszko, and Czartoryski), growing from 420k (1400AD) to 620k (1550AD) and 920k (1648AD), anchored by Ostroh (~12k, site of the Ostroh Academy and 1581AD Slavonic Bible), Lutsk (~11k), Kremenets (~8k), Dubno (~7,5k), and Volodymyr (~6k) [5]. Region 11: Podolian and Bratslav Voivodeships (39 620 km2): Spanning the Dniester and Southern Bug forest-steppe uplands, population rose from 350k (1400AD) to 540k (1550AD) and 910k (1648AD), led by the royal cliff fortress of Kamianets-Podilskyi (~15k), Vinnytsia (~9k), Bratslav (~8,5k), Bar (~7,5k), and Medzhybizh (~7k) [5][27].

Region 12: Right-Bank Kyiv Voivodeship and Cossack Regiments (56 764 km2): Covering the Right-Bank Dnieper, Teterev, Irpin, Ros, and Tyasmin basins, population surged from 280k (1400AD) and 440k (1550AD) to 920k in 1648AD as magnate *slobody* (granting 15-30 years of tax exemption) and Registered Cossack regiments proliferated around Kyiv (~18k in 1648AD, intellectual seat of Metropolitan Petro Mohyla's Collegium), Uman (~11k), Bila Tserkva (~10,5k), Korsun (~9,5k), and Chyhyryn (~8k, Bohdan Khmelnytsky's residence and first Hetmanate capital) [5][27].

Regions 13 and 14: Northern and Southern Left-Bank Hetmanate (55 782 km2; hetmanate_left_bank_south_poltava, 52 082 km2): In 1400AD, the Northern Left Bank (Region 13: Chernihiv, Nizhyn, Starodub, Putyvl, Baturyn, Hlukhiv) held 210k inhabitants and the Southern Left Bank (Region 14: Pereyaslav, Lubny, Myrhorod, Pryluky, Poltava, Hadiach) held 110k. After the 1618AD Truce of Deulino transferred Chernihiv-Seversk to Poland and the trans-Dnieper Wisniowiecki latifundium (*Wisniowiecczyzna*) centred on Lubny expanded from ~600 households in 1600AD to over 38 000 households (~230k inhabitants across 56 trans-Dnieper towns) by 1645AD, Region 13 reached 490k and Region 14 reached 460k in 1648AD (950k combined across the Left Bank) [5][27].

The Time of Troubles, the Deluge, and the Ukrainian Ruina (1598AD - 1687AD)

The Boris Godunov Famine (1601-1603AD), the Smuta, and the Post-Stolbovo Karelian Exodus

Following partial demographic recovery under Tsar Feodor Ivanovich and Boris Godunov (1584-1600AD) - which lifted Zamoskovny Moscow (Region 18) to 2 950k and Novgorod-Pskov (Region 19) to 1 040k in 1600AD - the Tsardom of Russia was struck by the Huaynaputina volcanic-winter famine of 1601-1603AD and the civil and international wars of the Time of Troubles (*Smuta*, 1604-1618AD) [1][50][51]. Post-*Smuta* inspection cadastres (*dozornye knigi* of 1614-1619AD) record sharp contractions across central and north-western Russia between 1600AD and 1615AD [1][21]:

Moscow and Zamoskovny Core (Region 18) fell by 22,7%, from 2 950k (1600AD) to 2 280k (1615AD), amid famine, the Bolotnikov rebellion, the Tushino camp, the 1610-1612AD Polish garrison in Moscow, and *Lisowczycy* light-cavalry raids. Novgorod and Pskov Five Pyatiny (Region 19) plunged by 31,7% (and by 47,0% relative to 1550AD), from 1 040k (1600AD) to 710k (1615AD), during Jacob De la Gardie's Swedish

occupation of Veliky Novgorod (1611-1617AD) and Gustavus Adolphus's siege of Pskov (1615AD). Olonets Karelia and Baltic Ingria (Region 20) dropped by 25,0%, from 240k (1600AD) to 180k (1615AD), mirroring Vyborg and Kexholm (Region 4, 95k to 80k). When the Treaty of Stolbovo (1617AD) ceded Baltic Ingria (*Ivangorod, Yam, Koporye, Noteburg*) alongside Kexholm County to Sweden, tens of thousands of Orthodox Karelians and Izhorians fled Swedish rule in the 1620s-1650sAD into the Bezhetskiy Verkh, Tver, Torzhok, and Tikhvin *uezds* (Regions 18-19) [38][40][41].

Smolensk and Seversk (Region 21) fell by 26,1% (690k to 510k) following the twenty-month Polish siege of Smolensk (1609-1611AD) and its cession at Deulino (1618AD), while Ryazan, Tula, and Oryol (Region 23) declined from 1 020k to 820k and the Belgorod and Sevsk Razryads (Region 24) dipped from 410k to 380k [51][52]. Conversely, sheltered Pomorye and Vologda (Region 22, 410k to 420k), Vyatka, Perm, and the Urals (Region 26, 390k to 405k), and the Eight Siberian Zones (Regions 31-38, 357k to 375k) escaped devastation and financed the 1611-1612AD Second Volunteer Army of Minin and Pozharsky from Nizhny Novgorod (Region 25) and Yaroslavl (Region 18) [1].

The Deluge (Krovavy Potop, 1654-1667AD) Across Lithuania and the Four Belarusian Regions

During the Russo-Polish War of 1654-1667AD (*Potop* / *Krovavy Potop*), both Core Lithuania (Region 3, 1 060k -> 640k, -39,6%) and the Four Belarusian GDL Regions (Regions 5-8) suffered one of the steepest demographic collapses in early modern Europe [9][13]. Comparing the 1648-1650AD and 1667AD *podymne* hearth-tax registers across all GDL *powiaty*, J. Morzy (1965) and A. M. Karpachev and P. G. Kozlovsky (1972) established that Belarus lost 48,8% of its total population, plunging from 2 500k in 1648AD to 1 280k at the Truce of Andrusovo (1667AD):

Region 5 (Polotsk and Vitebsk Voivodeships) fell by 51,8% (560k to 270k) amid Dvina sieges, the 1657-1658AD plague, and forced resettlements of skilled Belarusian townspeople (*belorustsy*) to Moscow [13]. Region 7 (Minsk, Orsha, and Mogilev Dnieper Corridor) dropped by 51,5% (680k to 330k) following the sack of Mstislavl (1654AD), the 1661AD Mogilev uprising, and continuous campaigning along the Smolensk-Orsha-Minsk road [9]. Region 8 (Brest-Litovsk and Pinsk-Turov-Mozyr Polesia) declined by 50,0% (520k to 260k), and Region 6 (Grodno and Novogrudok Voivodeships) contracted by 43,2% (740k to 420k) [9][13].

Although the four Belarusian regions partially recovered to 1 450k (1678AD) and 1 630k (1700AD), the Great Northern War (1700-1721AD) - marked by Russian, Swedish, Saxon, and Polish operations across Grodno, Polotsk, Lesnaya (1708AD), and Mogilev (burned in 1708AD), alongside the 1708-1711AD plague - pushed Belarus back to 1 430k in 1710AD and 1 550k in 1720AD (still 38,0% below its 1648AD peak) [9][16][49].

The Ukrainian Ruina (1657-1687AD) and the Shift to the Left-Bank and Sloboda Regiments

In Ukraine, the Khmelnytsky Uprising (1648-1657AD), the 1654AD Pereyaslav Agreement, and the thirty-year multi-sided conflict known as The Ruin (*Ruina*, 1657-1687AD) permanently shifted the demographic centre of gravity from Right-Bank Ukraine (Regions 10, 11, and 12) to the Left-Bank Hetmanate (Regions 13 and 14) and Sloboda Ukraine (Region 15) [5][27][53][54]:

Depopulation of Right-Bank Kyiv (Region 12), Podolia-Bratslav (Region 11), and Volhynia (Region 10): Between 1648AD and 1678AD, Right-Bank Kyiv Voivodeship and Cossack Regiments (Region 12) lost 60,9% of its population, plunging from 920k (1648AD) to 520k (1667AD), 360k (1678AD), and 320k (1700AD) after the Ottoman-Tatar Chyhyryn campaigns (1674-1678AD, destroying Uman and Chyhyryn), Hetman Ivan Samoilovych's 1678-1679AD *Velykyi Zhin* (forced trans-Dnieper evacuation to the Left Bank), and the 1686AD Eternal Peace buffer-zone clause (while the city of Kyiv itself remained under Russian garrison

rule on the Dnieper cliff at ~15k-18k) [5][53]. Simultaneously, Podolia and Bratslav (Region 11, ruled by the Ottoman Empire as the Eyalet of Kamanice in 1672-1699AD) fell by 58,2% (910k to 380k in 1678AD, recovering to 510k by 1720AD after the 1699AD Treaty of Karlowitz), Volhynia (Region 10) dropped from 920k to 580k (680k in 1720AD), and Galicia-Belz (Region 9) contracted from 1 750k to 1 380k (1 620k in 1720AD) [5].

Expansion of the Northern and Southern Left-Bank Hetmanate (Regions 13 and 14), absorbing waves of Right-Bank settlers, Northern Left-Bank Hetmanate (Region 13: Chernihiv, Nizhyn, Starodub, and Putyvl, plus the Hetman capitals of Baturyn 1669-1708AD and Hlukhiv 1708-1764AD) grew continuously from 490k (1648AD) to 560k (1667AD), 620k (1678AD), and 790k in 1720AD (+61,2%) [53]. Southern Left-Bank Hetmanate (Region 14: Pereyaslav, Lubny, Myrhorod, Pryluky, Hadiach, and Poltava Regiments) rose from 460k (1648AD) to 540k (1667AD), 610k (1678AD), and 760k in 1720AD (+65,2%), weathering the 1708-1709AD Swedish invasion and Battle of Poltava (June 1709AD) [5][53].

Boom of Sloboda Ukraine (Region 15: Kharkiv, Sumy, Okhtyrka, Ostrogozhsk, and Izium Regiments): Colonised by Ukrainian *cherkasy* migrants under Tsar Aleksey Mikhailovich's autonomous regimental charters (Ostrogozhsk 1652AD, Sumy 1652AD, Kharkiv 1654AD, Okhtyrka 1654AD, Izium 1681AD), Sloboda Ukraine (Region 15, 60 982 km2) surged from 110k (1600AD) and 160k (1648AD) to 250k (1667AD), 310k (1678AD), 380k (1700AD), and 400k in 1720AD (+150,0% above 1648AD) [17][54][55].

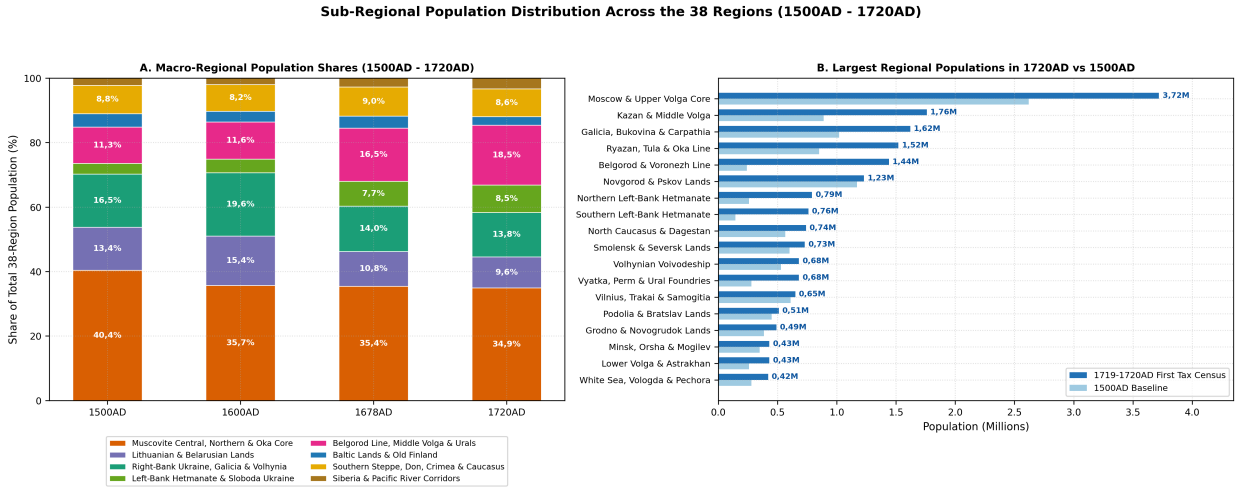


Figure 3. Regional percentage population shifts during major crisis and colonisation phases (Panel A) and Siberian population growth (1580AD - 1720AD, Panel B).

Frontier Razryads, Fortification Lines, and the Pontic-Caspian Steppe (Regions 16-17, 23-24, 28-30)

The Great Oka Abatis Line (Region 23) and the Belgorod-Sevsk Razryads (Region 24)

South of the Muscovite forest zone, the conversion of the forest-steppe chernozem from a Crimean-Nogai raiding ground (*Pole*) into the agrarian core of Russia unfolded across two frontier regions constructed from pre-modern *uezds* [18][52][56]:

Region 23: Ryazan, Tula, Oryol, and Great Oka Abatis Line (95 590 km2): Shielded by the felled-forest barriers, earthworks, and artillery gates of the Great Abatis Line (*Bolshaya Zasechnaya Cherta*, 1521-1566AD, rebuilt in 1638AD) running from Kozelsk, Belev, and Odoyev through Tula and Venev to Ryazan (*Pereyaslavl-Ryazansky*), Kolomna, Oryol (1566AD), and Yelets (1592AD), this region expanded from 680k (1400AD) to 980k (1550AD), recovered from its 1615AD *Smuta* dip (820k), and reached 1 080k (1648AD), 1 290k (1678AD), and 1 520k in 1720AD (9 persons/km2), with Tula (19k in 1720AD, home to the 1632AD Vinius ironworks and 1712AD Imperial Arms Plant) serving as Russia's premier weapons centre [7][18].

Region 24: Belgorod and Sevsk Razryads (148 778 km2): Blocking the Muravsky, Izyumsky, and Kalmiussky Tatar trails across the Kursk, Belgorod, Voronezh, Lipetsk, Kozlov, and Tambov chernozem *uezds*, the Military Chancellery (*Razryadny Prikaz*) founded Voronezh (1586AD), Livny (1586AD), Stary Oskol (1593AD), Belgorod (1596AD), and Kursk (1596AD), and in 1635-1658AD built the continuous 800 km Belgorod Fortified Line (*Belgorodskaya Cherta*) extending to Kozlov (1635AD) and Tambov (1636AD) [47][56]. Organised into the Belgorod Razryad (1658AD) and Sevsk Razryad, Region 24 grew more than seven-fold from 190k (1400AD) and 290k (1550AD) to 620k (1648AD), 980k (1678AD), and 1 440k in 1720AD (+132,3% above 1648AD) [7][57]. Its rural society was anchored by odnodvortsy ("single-homesteaders" descended from 17th-century *deti boyarskie*, cossacks, streltsy, and dragoons, exceeding 450k male souls by the 1720sAD), alongside expanding landlord serf estates and Peter I's Azov fleet shipyards at Voronezh (16k in 1720AD) and Lipetsk [18][58].

Frontier System / Regional Polity	Active Phase	Geographic Span	Principal Urban & Garrison Hubs	Dominant Social Estates	Regional Population Trajectory
Great Oka Abatis Line (Region 23)	1521-1638AD	600 km (Kozelsk - Tula - Kolomna - Ryazan)	Tula, Ryazan, Oryol, Kolomna, Yelets, Belev	Pomeshchiki, streltsy, gunsmiths & landlords' serfs	680k (1400AD) -> 1 080k (1648AD) -> 1 520k (1720AD)
Belgorod & Sevsk Razryads (Region 24)	1586-1658AD	800 km (Belgorod - Voronezh - Kozlov - Tambov)	Belgorod, Voronezh, Kursk, Tambov, Kozlov, Stary Oskol	Odnodvortsy (deti boyarskie), streltsy & serfs	190k (1400AD) -> 620k (1648AD) -> 1 440k (1720AD)
Kazan & Simbirsk-Penza Line (Region 25)	1552-1683AD	650 km (Saransk - Karsun - Simbirsk - Penza - Syzran)	Kazan, Nizhny Novgorod, Simbirsk, Saransk, Penza, Samara	Service Tatars/ Mordvins, yasak Chuvash/ Mari & serfs	780k (1400AD) -> 1 180k (1648AD) -> 1 760k (1720AD)
Sloboda Ukraine Regiments (Region 15)	1652-1681AD	530 km (Sumy - Okhtyrka - Kharkiv - Ostrogozhsk - Izium)	Kharkiv, Sumy, Okhtyrka, Izium, Ostrogozhsk, Valuyki	Ukrainian Cherkasy regimental cossacks & burghers	50k (1400AD) -> 160k (1648AD) -> 400k (1720AD)

(Continued from previous page)

Frontier System / Regional Polity	Active Phase	Geographic Span	Principal Urban & Garrison Hubs	Dominant Social Estates	Regional Population Trajectory
N. & S. Left-Bank Hetmanate (Regions 13-14)	1648-1720AD	Desna, Sula, Psyol & Vorskla regiment network	Baturyn, Hlukhiv, Nizhyn, Chernihiv, Poltava, Lubny	Registered Cossacks, pospolyti peasants & burghers	320k (1400AD) -> 950k (1648AD) -> 1 550k (1720AD)
Bashkir Four Darugas & Yaik (Region 27)	1574-1720AD	Belaya (Agidel), Ufa & Yaik river frontier	Ufa (1574AD), Birsk (1663AD), Menzelinsk, Yaitskiy Gorodok	Votchinniki Bashkirs (4 darugas), teptyari & Yaik Cossacks	180k (1400AD) -> 260k (1648AD) -> 340k (1720AD)

Table 4. Defensive fortification lines, frontier razryads, Cossack regimental polities, and population growth across Regions 13-15 and 23-27 (1400AD - 1720AD).

Zaporozhia, Crimea, the Lower Volga, the Don Host, and the North Caucasus (Regions 16-17, 28-30)

Beyond the settled forest-steppe lines lay five steppe, maritime, and mountain polities:

Region 16: Zaporozhian Sich and Pontic Wild Fields (193 859 km2; 60k in 1400AD to 130k in 1648AD to 180k in 1720AD): Centred on the island strongholds of the Zaporozhian Cossack Host below the Dnieper rapids (Khortytsia, Tomakivka, Bazavluk, Mykytyn Rih, Chortomlyk 1652-1709AD, Kamianka, and Oleshky), Polish river fort Kodak (1635AD), surrounding winter fishing-pastoral homesteads (*zymivnyky*), Ottoman Black Sea fortress Ochakiv (*Ozu*), and the Bakhmut salt-boiling forts [27][54]. Region 17: Crimean Khanate and Ottoman Eyalet of Kefe (32 002 km2; 210k in 1400AD to 360k in 1648AD to 370k in 1720AD): Divided after 1475AD between the direct Ottoman coastal province of Kefe (*Kaffa / Feodosia*) - holding ~65k-85k Muslim, Greek, Armenian, and Jewish inhabitants in 16th-century Ottoman *tahrir defterleri* across Kaffa, Sudak, Balaklava, Mangup, Kerch, and Taman [28][59] - and the Giray Crimean Khanate (~270k-290k inhabitants), anchored by Bakhchisarai (~20k-22k), Qarasubazar (*Karasubazar*) (~11k-12,5k), and Gozleve (*Yevpatoria*) (~10k) [28].

Region 28: Lower Volga Garrison Towns, Astrakhan, and Kalmyk Khanate (251 397 km2; 240k in 1400AD to 320k in 1648AD to 430k in 1720AD): Following Ivan IV's conquest of the Khanate of Astrakhan (1556AD), Muscovy secured the Volga-Caspian trade artery with Astrakhan (22k in 1720AD, hosting Russian, Tatar, Bukharan, Persian, Armenian, and Indian merchant compounds), Tsaritsyn (1589AD), and Saratov (1590AD), while Oirat pastoralists under Kho Orluk and Ayuka Khan (1669-1724AD) established the Kalmyk Khanate (~40k-50k *kibitki* or ~200k-250k nomads) across the surrounding Caspian-Volga steppes [7][48][60].

Region 29: Don Cossack Host and Ottoman/Petrine Azov (169 534 km2; 90k in 1400AD to 165k in 1648AD to 240k in 1700AD to 230k in 1710AD to 250k in 1720AD): Along the Lower and Middle Don, Seversky Donets, Khopyor, and Medveditsa, the autonomous Don Cossack Host established over 120 fortified *gorodki* governed from Razdory and Cherkassk (capital from 1644AD, ~8,5k in 1720AD, plus Mikhaylovskiy Gorodok on the Khopyor), contesting Ottoman Azov (held by the Cossacks in 1637-1642AD and by Peter I in 1696-1711AD alongside newly built Taganrog 1698AD) and experiencing a brief demographic dip during

the Bulavin Rebellion of 1707-1708AD [47][61].

Region 30: Circassia, Kabarda, Vainakh Societies, Dagestan Shamkhalate, and Terskiy Gorod (263 210 km2; 520k in 1400AD to 665k in 1648AD to 740k in 1720AD): Encompassing Adyghe/Circassian societies, Greater and Lesser Kabarda (allied with Moscow from 1557AD), Vainakh (*Chechen* and *Ingush*) highland tower communities and piedmont settlements [62], the Russian Terek delta fortress of Terskiy Gorod (*Terki*, 1588AD) with the Grebensky Cossacks, and the polities of Dagestan (Derbent, the Kumyk Shamkhalate of Tarki, Endirey, Kumukh, the Avar Khanate, and mountain *jamaat* unions) [63].

Eastwards Colonisation: Kazan, the Urals, Bashkiria, and the Eight Siberian Zones (Regions 25-27, 31-38)

The Middle Volga (Region 25), Perm-Stroganov Urals (Region 26), and Bashkir Four Darugas (Region 27)

East of the Oka and Sura rivers, the conquest of the Khanate of Kazan (1552AD) integrated the Middle Volga, Kama, and Ural basins into the Russian state across three regions:

Region 25: Former Khanate of Kazan, Nizhny Novgorod, Chuvash-Mari Lands, and Simbirsk-Penza Line (237 582 km2; 780k in 1400AD to 940k in 1550AD to 1 180k in 1648AD to 1 760k in 1720AD): Combining Nizhny Novgorod (15k in 1720AD, adjacent to the Makaryev Fair), Arzamas, Kasimov, and Temnikov with the former Kazan Khanate (Kazan 30k in 1720AD, Sviyazhsk 1551AD, Cheboksary 1555AD, Samara 1586AD) and the Simbirsk-Kama (1648-1656AD) and Penza-Syzran (1663-1683AD) fortification lines (Saransk 1641AD, Simbirsk 1648AD, Penza 1663AD, Syzran 1683AD), populated by Russian peasants alongside *yasak* and service (*sluzhilye*) Volga Tatars, Chuvash, Mordvins, Mari, and Udmurts [7][64][65][66].

Region 26: Great Perm, Vyatka, Solikamsk-Stroganov Salt Lands, and Early Demidov Ural Foundries (479 789 km2; 220k in 1400AD to 330k in 1550AD to 480k in 1648AD to 680k in 1720AD): Linking the *chernososhnye* grain basin of Vyatka (*Khlynov*) with Perm Velikaya (Cherdyn, Solikamsk 9,8k in 1720AD, and the Stroganov Kama-Chusovaya salt domains), the Sylva agricultural hub of Kungur (1648/1663AD), and Nikita Demidov's Petrine blast-furnace ironworks in the Middle Urals (Nevyansk 1701AD, Kamensk 1701AD, Alapaevsk 1704AD) [7][8].

Region 27: Bashkir Four Darugas (Ufa Ostrog) and Yaik Cossack Steppe Frontier (484 827 km2; 180k in 1400AD to 215k in 1550AD to 260k in 1648AD to 340k in 1720AD): Administered after Ufa's foundation (1574AD) through the four Bashkir *Darugas* (Kazan, Nogai, Osinskaya, and Siberian), where hereditary Bashkir *votchinniki* clans defended their patrimonial land rights in the uprisings of 1662-1664AD, 1681-1684AD, and 1704-1711AD alongside Russian fortresses (Birsk 1663AD, Menzelinsk), *teptyari* migrants, and the Yaik Cossack Host at Yaitskiy Gorodok [7][57].

The Eight Disaggregated Siberian Portage, Ploughland, Steppe, and Yasak Zones (Regions 31-38: 1582AD - 1721AD)

East of the Ural divide, Russian servicemen (*sluzhilye lyudi*), fur trappers (*promyshlenniki*), and state plough peasants (*pashennye krestyane*) advanced 6 000 kilometres from Yermak's 1582AD capture of Qashliq (*Isker*) to Okhotsk (1647AD), the Bering Strait (1648AD), the Amur (1651-1689AD), and Kamchatka (1697AD) [6][20][67][68]. Because early modern Siberian settlement followed east-west river-portage highways (*voloki*), southern fortified ploughlands (*gosudareva desyatinnaya pashnya*), unsubjugated nomadic steppe borderlands, and boreal/Arctic *yasak* fur districts, our partition disaggregates Northern Asia (12 806 014 km2 across Regions 31-38, rising from 300k in 1400AD and 338k in 1580AD to 435k in

1648AD, 545k in 1678AD, and 760k in 1720AD) into eight functional zones (Panel B of Figure 1, Panel B of Figure 3, and Table 5):

Region 31: Pomor-Mangazeya Maritime and Taz-Turukhansk Arctic Portage Route (1 760 905 km2; 16k in 1400AD to 19k in 1580AD to 28k peak in 1615AD to 26k in 1648AD to 18k in 1720AD): Pomor mariners sailing *kochi* across the Kara Sea and the Yamal portage founded Obdorsk (1595AD), Mangazeya (1601AD) on the Taz River, and Turukhansk (1607AD), lifting seasonal and indigenous Nenets, Enets, Nganasan, and Selkup numbers to 28k in 1615AD before Tsar Mikhail Romanov's 1619AD decree closing the Mangazeya Sea Route, sable depletion, and the 1642AD Mangazeya fire relocated administration to Novaya Mangazeya (*Turukhansk*, 1672AD) and reduced Region 31 to 18k by 1720AD [6][67][69].

Region 32: Tobolsk-Irtysh-Middle Ob River Portage Highway (871 344 km2; 42k in 1400AD to 49k in 1580AD to 88k in 1648AD to 118k in 1678AD to 158k in 1720AD): Controlled via the Babylka Ural Portage (1598AD) at Verkhoturye (1598AD), Turinsk (1600AD), Pelym (1593AD), Tyumen (1586AD), and the Siberian capital Tobolsk (1587AD, 15k in 1720AD), and extending down the Lower Irtysh (Berezov 1593AD) and Middle Ob (Surgut 1594AD, Narym 1596AD, Ketsky Ostrog 1602AD) across Khanty, Mansi, and Siberian Tatar *yasak* volosts [7][70][71][72].

Region 33: Southern Tobol-Ishim-Tara-Tomsk-Kuznetsk Fortified Agricultural Frontier (359 192 km2; 38k in 1400AD to 47k in 1580AD to 86k in 1648AD to 145k in 1678AD to 254k in 1720AD): Stretching along the forest-steppe chernozem from the Iset-Tobol *slobody* (Shadrinsk 1662AD, Kurgan 1679AD), Tara (1594AD), and Omsk (1716AD) to Tomsk (1604AD, 8,5k in 1720AD) and Kuznetsk (1618AD), Region 33 became the primary demographic engine of Siberia, expanding more than five-fold from 47k (1580AD) to 254k in 1720AD and housing over 60% of Northern Asia's Russian plough peasantry [20][73][74].

Region 34: Nomadic Baraba-Altai-Minusinsk-Tuva Steppe Borderlands (821 185 km2; 74k in 1400AD to 81k in 1580AD to 84k in 1648AD to 80k in 1678AD to 100k in 1720AD): A contested zone of dual tribute (*dvoedanstvo*) inhabited by Baraba Tatars, Teleuts, Shors, Tuvans, and the four Yenisei Kirghiz *ulusy* of Minusinsk until the 1703AD Dzungar relocation of the Kirghiz elite enabled Russian construction of Abakansky Ostrog (1707AD) and Biysky Ostrog (1709AD) [6][73]. Region 35: Ket-Yeniseysk-Angara-Ilimsk-Irkutsk River Portage Spine (1 931 504 km2; 28k in 1400AD to 32k in 1580AD to 44k in 1648AD to 66k in 1678AD to 98k in 1720AD): Linking the Makovskiy Ket-Yenisei portage (1618AD) through Yeniseysk (1619AD, 8k in 1720AD) and Krasnoyarsk (1628AD) up the Angara to Ilimsk (1630AD, Lena Portage) and Irkutsk (1661AD, 6,5k in 1720AD) among Western Buryats, Kets, and Evenks [6][7][67].

Region 36: Transbaikal Dauria and Upper Amur Frontier (1 753 934 km2; 34k in 1400AD to 38k in 1580AD to 40k in 1648AD to 48k in 1678AD to 62k in 1720AD): Anchored by Nerchinsk (1658AD), Selenginsk (1665AD), and Udinsk (1666AD) among Khori-Selenga Buryats, Evenks, and Daurs, after Qing sieges of Albazin (1651-1689AD) and the Treaty of Nerchinsk (1689AD) redirected Russian settlement into Nerchinsk silver-lead mining (1704AD) and Kiakhta-Selenga caravan trade [6][75][76]. Region 37: Central Lena-Vilyuy-Aldan Basin (1 373 412 km2; 38k in 1400AD to 42k in 1580AD to 42k in 1648AD to 44k in 1678AD to 47k in 1720AD): Centred on Yakutsk (1632AD), Vilyuysk (1634AD), and Olekminsk (1635AD) in the thermokarst *alaas* meadows of the Yakut (Sakha) cattle- and horse-breeding pastoralists (~28,5k-35k Sakha across 35 *yasak* volosts plus Evenks and ~4k-5k Russians by 1720AD) [6][77].

Region 38: Northeast Arctic and Pacific Yasak Frontier (3 924 538 km2; 30k in 1400AD to 30k in 1580AD to 25k in 1648AD to 25k in 1678AD to 23k in 1720AD): Spanning Okhotsk (1647AD), Anadyrsk (1649AD), and Nizhnekamchatsk (1697/1703AD), where virgin-soil smallpox epidemics (1651-1653AD, 1669AD, 1691-1694AD) and frontier conflict reduced Yukaghir, Itelmen, Koryak, Even, and Chukchi populations from 30k to 23k [6][76][78].

Siberian Functional Zone (Reg. 31-38)	Area	Principal Ostrogi, Portages & Indigenous Ulusy	1580AD	1615AD	1648AD	1678AD	1720AD
31. Pomor-Mangazeya Maritime & Taz Portage	1 760 905 km2	Obdorsk (1595AD), Mangazeya (1601AD; closed 1619AD), Turukhansk (1607/ 1672AD)	19k	28k	26k	19k	18k
32. Tobolsk-Irtysh-Middle Ob Portage Highway	871 344 km2	Tyumen (1586AD), Tobolsk (1587AD), Berezov (1593AD), Surgut (1594AD), Verkhoturye (1598AD)	49k	64k	88k	118k	158k
33. S. Tobol-Tara-Tomsk-Kuznetsk Ploughland	359 192 km2	Tara (1594AD), Tomsk (1604AD), Kuznetsk (1618AD), Shadrinsk (1662AD), Omsk (1716AD)	47k	60k	86k	145k	254k
34. Nomadic Baraba-Altai-Minusinsk-Tuva Steppe	821 185 km2	Baraba & Teleut ulusy, Hongorai Kirghiz; Abakansky (1707AD), Biysky (1709AD)	81k	82k	84k	80k	100k
35. Ket-Yeniseysk-Angara-Ilimsk-Irkutsk Spine	1 931 504 km2	Makovskiy (1618AD), Yeniseysk (1619AD), Krasnoyarsk (1628AD), Ilimsk (1630AD), Irkutsk (1661AD)	32k	34k	44k	66k	98k
36. Transbaikal Dauria & Upper Amur Frontier	1 753 934 km2	Albazin (1651-1689AD), Nerchinsk (1658AD), Selenginsk (1665AD), Udinsk (1666AD)	38k	39k	40k	48k	62k
37. Central Lena-Vilyuy-Aldan (Yakutsk Alaas)	1 373 412 km2	Yakutsk (1632AD), Vilyuysk (1634AD), Olekminsk (1635AD), Amginskaya Sloboda (1652AD)	42k	41k	42k	44k	47k

(Continued from previous page)

Siberian Functional Zone (Reg. 31-38)	Area	Principal Ostrogi, Portages & Indigenous Ulusy	1580AD	1615AD	1648AD	1678AD	1720AD
38. NE Arctic & Pacific Yasak (Kolyma-Kamchatka)	3 924 538 km2	Verkhoyansk (1638AD), Srednekolymsk (1643AD), Okhotsk (1647AD), Anadyrsk (1649AD), Nizhnekamchatsk (1697AD)	30k	27k	25k	25k	23k
Total Siberia & Far East (Regions 31-38)	12 806 014 km2	Over 50 ostrogi, slobody, portage stations, and goroda across Northern Asia	338k (2k RU)	375k (25k RU)	435k (85k RU)	545k (165k RU)	760k (365k RU)

Table 5. Land areas, principal towns, and benchmark population series across the eight Siberian river, steppe, and frontier zones (1580AD - 1720AD).

Petrine Reforms, Urban Hierarchy, and the 1720/1721AD Imperial Splice

Saint Petersburg, the First Poll-Tax Revision, and the 1720AD Urban Hierarchy

The reign of Peter I (1682/1689-1725AD) and the Great Northern War (1700-1721AD) transformed both the administrative structure and the urban hierarchy of the realm [7][8][24]. After capturing Swedish Noteburg (Shlisselburg) and Nyenskans on the Neva in Region 20 (muscovy_karelia_ingria_st_petersburg), Peter I founded Saint Petersburg (May 1703AD) and the naval fortress of Kronstadt (1704AD), reinforced in 1710AD by the conquest of Vyborg, Kexholm (Region 4), Riga, Pernau, Arensburg, and Reval (Region 1). Driven by annual conscription of 20k-40k *rabotnye lyudi* labourers and the mandatory transfer of noble families, foreign-trade merchants, and central colleges (1712-1718AD), Saint Petersburg expanded from ~5k in 1703AD to ~18k in 1710AD and 40k in 1720AD, lifting Region 20 from 270k (1700AD) to 350k (1720AD) and making the new Baltic capital the second largest city in the Empire behind Moscow (Region 18, 160k) [7][24].

Following the 1708-1710AD guberniya reform (creating Moscow, Ingermanland/Saint Petersburg, Kiev, Smolensk, Arkhangelogorod, Kazan, Azov/Voronezh, and Siberia, joined after 1710-1713AD by Riga/Livland and Reval/Estland) and the 1710AD household audit - which exposed wartime recruitment losses, plague mortality in the western provinces, and widespread household concealment (*utayka dvorov*, reflected in the aggregate dip from 22 060k in 1700AD to 21 549k in 1710AD) - the Senate enacted the First Poll-Tax Revision (1718-1721AD) [7][8][22]. By 1720AD, the urban hierarchy across all thirty-eight regions (Figure 4) encompassed distinct macro-regional tiers:

In Russia and Siberia (Regions 18-38, 14 370k total): Behind Moscow (Region 18, 160k) and Saint Petersburg (Region 20, 40k), the leading centres were Kazan (Region 25, 30k), Yaroslavl (Region 18, 26k),

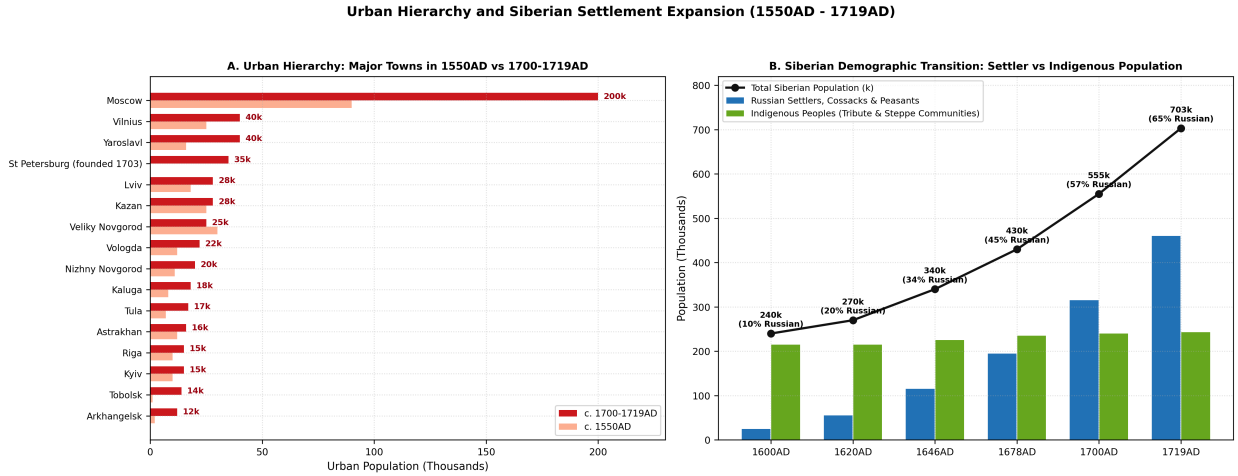


Figure 4. Largest urban centres in 1500AD versus 1648AD (Panel A) and leading cities at the First Poll-Tax Revision of 1720AD (Panel B).

Astrakhan (Region 28, 22k), Kaluga (Region 18, 20k), Tula (Region 23, 19k), Voronezh (Region 24, 16k), Nizhny Novgorod (Region 25, 15k), Tobolsk (Region 32, 15k), Pskov (Region 19, 15k), Veliky Novgorod (Region 19, 14,5k), Kostroma (Region 18, 14,5k), Vologda (Region 22, 14k), Smolensk (Region 21, 14k), Kursk (Region 24, 12,5k), Kolomna (Region 23, 12,5k), Belgorod (Region 24, 11k), and Oryol (Region 23, 11k) [7][24].

In Ukraine and Crimea (Regions 9-17, 5 650k total): Lviv (Region 9, 25k), Bakhchisarai (Region 17, 20k), and Kaffa (Region 17, 19k) led the western and southern rims, while Kyiv (Region 12, 18k), Nizhyn (Region 13, 14k), Chernihiv (Region 13, 13,5k), Hlukhiv (Region 13, 12,5k), Poltava (Region 14, 12k), Starodub (Region 13, 11,5k), Pereyaslav (Region 14, 11,5k), Kamianets-Podilskyi (Region 11, 11,5k), and Kharkiv (Region 15, 11k) anchored the Hetmanate, Sloboda Ukraine, and Podolia [5][28][53].

In the Ostsee Baltics, Courland, Core Lithuania, Belarus, and Old Finland (Regions 1-8, 2 810k total): Recovering from the 1708-1711AD plague and Great Northern War sieges, Mogilev (Region 7, 18k) and Vilnius (Region 3, 13,5k, down from 28k in 1648AD) led the Lithuanian-Belarusian interior alongside Slutsk (Region 7, 9,5k), Polotsk (Region 5, 9k), Vitebsk (Region 5, 8,5k), Grodno (Region 6, 8,5k), Minsk (Region 7, 8k), Brest-Litovsk (Region 8, 8k), and Kaunas (Region 3, 6,5k), while Riga (Region 1, 9,5k), Reval / Tallinn (Region 1, 6,8k), Mitau / Jelgava (Region 2, 4,2k), Narva (Region 1, 4,2k), Vyborg (Region 4, 3,5k), Libau (Region 2, 3,2k), and Dorpat / Tartu (Region 1, 2,8k) anchored the Nystad Baltic littoral [3][9][10][35].

Continuity and Spatial Envelope Splice at 1720/1721AD into Early Imperial Russia

At the Treaty of Nystad (30 August / 10 September 1721AD), Peter I assumed the title of Emperor (*Imperator Vserossiyskiy*), closing the tsardom_of_russia epoch (1400AD - 1721AD, 38 regions, 18 119 929 km2, 448 088 pixels; 22 830k in 1720AD) and splicing directly into the imperial_russia_pre_1815 epoch (1721AD - 1815AD, 40 regions, 18 496 636 km2, 457 996 pixels; 23 888k in 1721AD) [8][57]. Across the newly annexed Nystad and Polish-Lithuanian border provinces, regional populations splice one-to-one: Region 1 (300k in 1720AD) partitions cleanly into Estland Governorate (105k in 1721AD) and Livland Governorate (195k in 1721AD); Region 2 (200k) aligns with Courland (200k); Region 3 (650k) aligns with Vilna and Samogitia (655k); and Region 4 (110k) continues directly as Old Finland / Vyborg Governorate (110k). The net difference of +1 058k (+376 707 km2, +9 908 pixels) between the 1720AD Tsardom terminal total (22 830k) and the 1721AD Early Imperial baseline (23 888k) reflects two transparent components: first, the +590k

spatial envelope expansion adding the Grand Duchy of Finland (annexed 1809AD at Fredrikshamn, 333 899 km2, 390k in 1721AD) and Bessarabia / Budjak (annexed 1812AD at Bucharest, 47 077 km2, 200k in 1721AD) so the 1721-1815AD partition holds a fixed spatial denominator; and second, +468k (+2,05%) across the continuous 38-region core from post-Nystad demographic recovery and the military revision *popolnenie* audits of 1721-1724AD [7][8].

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